

A. A Binary System

A-1. Assume a_1 and a_2 , are respectively, the distances of M_1 and M_2 from the center of mass:

$$\begin{cases} M_1 a_1 = M_2 a_2 \\ a_1 + a_2 = a \end{cases} \rightarrow a_1 = \frac{M_2}{M} a, a_2 = \frac{M_1}{M} a : M = M_1 + M_2$$

In the rotating coordinate system, a centrifugal potential has to be added to the gravitational potential of the two masses:

$$U = -\frac{1}{2} \omega^2 r^2 \quad \omega = \sqrt{\frac{GM}{a^3}}$$

$$\varphi(x, y) = -\frac{GM_1}{\sqrt{(x + a_1)^2 + y^2}} - \frac{GM_2}{\sqrt{(x - a_2)^2 + y^2}} - \frac{1}{2} \omega^2 (x^2 + y^2)$$

$$\varphi(x, y) = -\frac{GM_1}{\sqrt{\left(x + \frac{M_2}{M} a\right)^2 + y^2}} - \frac{GM_2}{\sqrt{\left(x - \frac{M_1}{M} a\right)^2 + y^2}} - \frac{1}{2} \frac{GM}{a^3} (x^2 + y^2)$$

A-1 (1.0 pt)

$$\varphi(x, y) = -\frac{GM_1}{\sqrt{\left(x + \frac{M_2}{(M_1 + M_2)} a\right)^2 + y^2}} - \frac{GM_2}{\sqrt{\left(x - \frac{M_1}{(M_1 + M_2)} a\right)^2 + y^2}} - \frac{1}{2} \frac{G(M_1 + M_2)}{a^3} (x^2 + y^2)$$

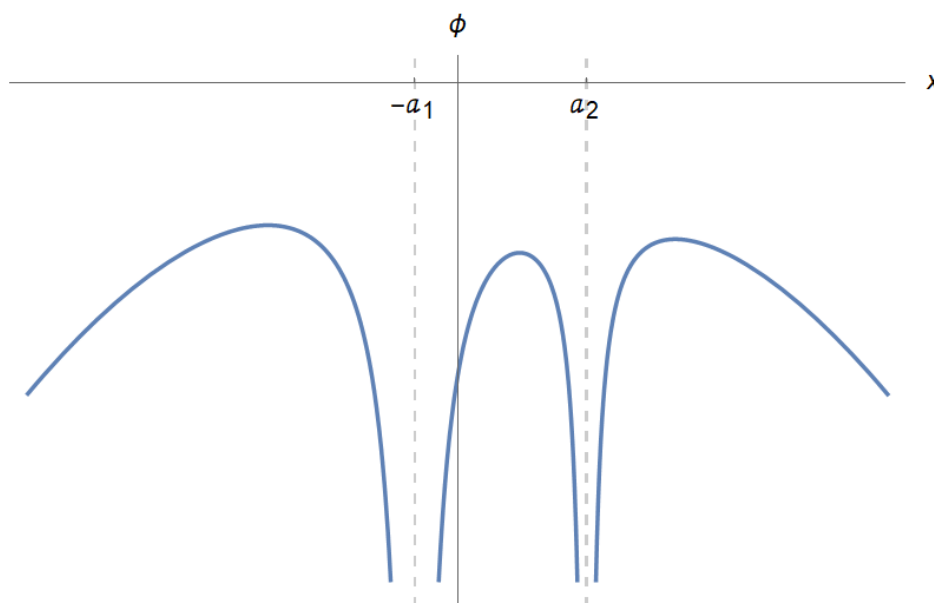
A-2. We set $y = 0$ in the previous equation, and obtain:

$$\varphi(x, 0) = -\frac{GM_1}{\left|x + \frac{M_2}{M} a\right|} - \frac{GM_2}{\left|x - \frac{M_1}{M} a\right|} - \frac{1}{2} \frac{GM}{a^3} x^2$$

We draw the diagram noting that:

1. The function has asymptotes at $x = -a_1$ and $x = a_2$, and it tends to $-\infty$ at both sides of these asymptotes.
2. The function has three maxima which are called Lagrange points.
3. The function goes to $-\infty$ for $x \rightarrow \pm\infty$

A-2 (0.7 pt)



A-3. Let $\bar{x} = x/a$, and denote the Lagrange point in the middle (between $\bar{x} = 0$ and $\bar{x} = 0.75$) by $\bar{x}_0$, we have $\frac{d\phi}{d\bar{x}}(\bar{x}_0) = 0$. Using the given ratios:

$$\phi(\bar{x}, 0) = \frac{GM}{a} \left[-\frac{\frac{3}{4}}{\left(\bar{x} + \frac{1}{4}\right)} + \frac{\frac{1}{4}}{\left(\bar{x} - \frac{3}{4}\right)} - \frac{1}{2} \bar{x}^2 \right]$$

Let $f(\bar{x}) = \frac{a}{GM} \frac{d\phi}{d\bar{x}}$, then we have to solve for $f(\bar{x}_0) = 0$. We have $f(0) > 0$ and $f(0.5) < 0$, so the answer lies between 0 and 0.5. For the midpoint, we have $f(\bar{x}_0 = 0.25) > 0$ so $0.25 < \bar{x}_0 < 0.5$, so by trial and error:

$$\begin{aligned} \begin{cases} f(0) > 0 \\ f(0.5) < 0 \end{cases} &\rightarrow f(0.25) > 0 \rightarrow 0.25 < \bar{x}_0 < 0.5 \rightarrow f(0.375) < 0 \rightarrow \dots \rightarrow 0.358 < \bar{x}_0 < 0.361 \\ &\rightarrow f(0.360) > 0 \rightarrow 0.25 < \bar{x}_0 < 0.361 \rightarrow \frac{x_0}{a} = \bar{x}_0 \approx 0.36 \end{aligned}$$

So, up to two significant figures the answer is 0.36.

A-3 (0.5 pt)

$$\frac{x_0}{a} = 0.36$$

A-4. The angular momentum of the system is:

$$J = \mu a V = \mu a^2 \omega = \frac{M_1 M_2}{M} a^2 \sqrt{\frac{GM}{a^3}} = \sqrt{\frac{GM_1^2 M_2^2}{M}} a,$$

where μ is the reduced mass and V is the relative velocity of the two point masses. Taking the logarithm of both sides we'll have:

$$\ln J = \frac{1}{2} \left[\ln \frac{G}{M} + 2 \ln M_1 + 2 \ln M_2 + \ln a \right]$$

For slowly-varying quantities we'll obtain:

$$\frac{\dot{J}}{J} = \frac{\dot{M}_1}{M_1} + \frac{\dot{M}_2}{M_2} + \frac{1}{2} \frac{\dot{a}}{a}$$

because the total mass is a constant and $\dot{M}_1 + \dot{M}_2 = 0$; therefore:

$$\frac{\dot{a}}{a} = -2 \frac{\dot{M}_1}{M_1} \left(1 - \frac{M_1}{M_2} \right) \rightarrow \dot{a} = -2\beta a \left(\frac{1}{M_1} - \frac{1}{M_2} \right)$$

For the period we'll have:

$$P = 2\pi \sqrt{\frac{a^3}{GM}} \rightarrow \frac{\dot{P}}{P} = \frac{3}{2} \frac{\dot{a}}{a} = -3 \frac{\dot{M}_1}{M_1} \left(1 - \frac{M_1}{M_2} \right) \rightarrow \dot{P} = -6\pi \sqrt{\frac{a^3}{GM}} \beta \left(\frac{1}{M_1} - \frac{1}{M_2} \right)$$

A-4 (0.6 pt)

$$\dot{a} = -2\beta a \left(\frac{1}{M_1} - \frac{1}{M_2} \right)$$

$$\dot{P} = -6\pi \sqrt{\frac{a^3}{GM}} \beta \left(\frac{1}{M_1} - \frac{1}{M_2} \right)$$

A-5. In an infinitesimally thin ring with an inner radius of r and an outer radius $r + dr$, energy is leaving at a rate of $-\frac{GM_1\beta}{2r}$ and entering at a rate $-\frac{GM_1\beta}{2r} + \frac{GM_1\beta}{2r^2} dr$. For the ring to stay in equilibrium, the excess energy of $\frac{GM_1\beta}{2r} dr$ per unit time must leave the system as radiation, so:

$$dP = \frac{GM_1\beta}{2r^2} dr = \sigma T^4 2(2\pi r dr) = 4\pi\sigma T^4 dr \rightarrow T = \left(\frac{GM_1\beta}{8\pi\sigma r^3} \right)^{\frac{1}{4}}$$

A-5 (1.0 pt)

$$T = \left(\frac{GM_1\beta}{8\pi\sigma r^3} \right)^{\frac{1}{4}}$$

A-6. From $P = 2\pi \sqrt{\frac{a^3}{GM}}$ we'll have:

$$a = \left[\frac{P^2 G (M_S + M_{NS})}{4\pi^2} \right]^{\frac{1}{3}}$$

Using the result of Part A.5, the temperature is:

$$T = \left(\frac{GM_{NS}\beta}{8\pi\sigma r^3} \right)^{\frac{1}{4}} = \left(\frac{500\pi M_{NS}\beta}{\sigma P^2 (M_S + M_{NS})} \right)^{\frac{1}{4}} = 9 \times 10^3 K$$

A-6 (0.5 pt)

$$T = 9 \times 10^9 K \times 10^2$$

A.7. For the system to remain bounded, the total mechanical energy of the system must be negative:

$$E' = \frac{1}{2} \mu' v'^2 - \frac{GM'_1 M_2}{a} < 0 \rightarrow v' < \sqrt{\frac{2G(M'_1 + M_2)}{a}}$$

For an isotropic explosion, we would have $v' = v = \sqrt{\frac{GM}{a}}$ therefore:

$$\sqrt{\frac{G(M_1 + M_2)}{a}} < \sqrt{\frac{2G(M'_1 + M_2)}{a}}$$

and:

$$\frac{M_1 - M_2}{2} < M'_1$$

A-7 (0.7 pt)

$$v'_{\max} = \sqrt{\frac{2G(M'_1 + M_2)}{a}}$$

$$M'_{1\min} = \frac{M_1 - M_2}{2}$$

B. Analysis of the stability of a star

B-1. Using Newton's law of gravity:

$$g = -\frac{4\pi G \int_0^r r'^2 \rho dr'}{r^2} \stackrel{\rho \cong \rho_c}{=} -\frac{4\pi G \rho_c r}{3}$$

B-1 (0.2 pt)

$$g = -\frac{4\pi G \rho_c r}{3}$$

B-2. Balance of forces for a differential element of volume with a surface area of A and thickness Δr between radii r and $r + \Delta r$ is as follows:

$$\vec{F} = -\frac{GM(\vec{r})\rho}{r^2} A \Delta r - \Delta p A = 0$$

in which $M(r)$ is the mass of the part of the star confined within the radius r . As Δr is small, we can write:

$$\frac{G\rho}{r^2} \left(\int 4\pi r'^2 \rho(r') dr' \right) = -\frac{dp(r)}{dr} = -K\gamma \rho^{\gamma-1} \frac{d\rho}{dr}$$

Multiplying both sides of the equation by $\frac{r^2}{4\pi G\rho}$ and taking the derivative once again, we get:

$$\frac{d}{dr} \left[r^2 \rho^{\gamma-2} \frac{d\rho}{dr} \right] + \frac{4\pi G r^2}{K\gamma} \rho(r) = 0$$

B-2 (0.6 pt)

$$h_1(\rho, r) = r^2 \rho^{\gamma-2}$$

$$h_2(r) = \frac{4\pi G r^2}{K\gamma}$$

B-3.

$$[\rho_c] = ML^{-3}, \quad [p_c] = ML^{-1}T^{-2}, \quad [G] = M^{-1}L^3T^{-2}$$

$$[G^l p_c^m \rho_c^n] = (M^{-1}L^3T^{-2})^l (ML^{-1}T^{-2})^m (ML^{-3})^n = L$$

$$\begin{cases} -l + n + m = 0 \\ 3l - 3n - m = 1 \\ -2l - 2m = 0 \end{cases} \rightarrow \begin{cases} l = -\frac{1}{2} \\ m = \frac{1}{2} \\ n = -1 \end{cases} \rightarrow r_0 = G^{-\frac{1}{2}} p_c^{\frac{1}{2}} \rho_c^{-1}$$

B-3 (0.4 pt)

$$r_0 = G^{-\frac{1}{2}} p_c^{\frac{1}{2}} \rho_c^{-1}$$

B-4.

$$\frac{K\gamma\rho_c^{\gamma-1}}{4\pi Gr_0^2 x^2} \frac{d}{dx} \left[x^2 u^{\gamma-2} \frac{du}{dx} \right] = -\rho_c u(r)$$

$$\frac{K\gamma\rho_c^{\gamma-2}}{4\pi Gr_0^2 x^2} \frac{d}{dx} \left[x^2 u^{\gamma-2} \frac{du}{dx} \right] = \frac{\gamma}{4\pi x^2} \frac{d}{dx} \left[x^2 u^{\gamma-2} \frac{du}{dx} \right] = -u$$

$$\frac{d}{dx} \left[x^2 u^{\gamma-2} \frac{du}{dx} \right] + \frac{4\pi x^2}{\gamma} u = 0$$

B-4 (0.3 pt)

$$A_1(u, x) = x^2 u^{\gamma-2}$$

$$A_2(x) = \frac{4\pi x^2}{\gamma}$$

B-5.

$$\gamma = 2 \rightarrow \frac{d}{dx} \left[x^2 \frac{du}{dx} \right] = -2\pi x^2 u(x) \rightarrow f''(x) = -2\pi f(x) \rightarrow f(x) = \frac{\sin(\sqrt{2\pi}x)}{\sqrt{2\pi}}$$

B-5 (0.6 pt)

$$f(x) = \frac{\sin(\sqrt{2\pi}x)}{\sqrt{2\pi}}$$

B.6.

$$\frac{d^2 u}{dx^2} + \frac{(\gamma-2)}{u} \left(\frac{du}{dx} \right)^2 + \frac{2}{x} \left(\frac{du}{dx} \right) + \frac{4\pi}{\gamma} u^{3-\gamma} = 0$$

$$u'(0) = 0, \quad \lim_{x \rightarrow 0} \frac{u'(x)}{x} = u''(0)$$

$$u''(0) + 2u''(0) + \frac{4\pi}{\gamma} = 0 \rightarrow \gamma = -\frac{4\pi}{3u''(0)}$$

$$\gamma \sim [1.64, 1.70]$$

B-6 (0.8 pt)

$$\gamma = [1.64, 1.70]$$

B.7.

$$M(r) = \int_0^{\tilde{r}(r,t)} 4\pi r'^2 \tilde{\rho}(r', t) dr' = \int_0^r 4\pi r'^2 \rho(r') dr'$$

$$4\pi r^2 \rho(r) = 4\pi \tilde{r}^2 \tilde{\rho}(\tilde{r}, t) \frac{\partial \tilde{r}}{\partial r} \rightarrow \frac{\tilde{\rho}}{\rho} = \frac{r^2}{\tilde{r}^2} \left(\frac{\partial \tilde{r}}{\partial r} \right)^{-1} = (1 + \epsilon)^{-3} \cong 1 - 3\epsilon$$

$$\frac{\tilde{g}}{g} = \frac{\frac{GM}{\tilde{r}^2}}{\frac{GM}{r^2}} = \frac{1}{\tilde{r}^2} = (1 + \epsilon)^{-2} \cong 1 - 2\epsilon$$

B-7 (0.9 pt)

$$\tilde{g} \simeq g(1 - 2\epsilon)$$

$$\tilde{\rho} \simeq \rho(1 - 3\epsilon)$$

B-8. we have

$$\frac{\partial \tilde{p}}{\partial \tilde{r}} = \tilde{\rho}(\tilde{g} - \ddot{\tilde{r}})$$

And

$$\tilde{p} = K \tilde{\rho}^\gamma$$

So:

$$\ddot{\tilde{r}} = \tilde{g} - \frac{\left(\frac{\partial \tilde{p}}{\partial \tilde{r}} \right)}{\tilde{\rho}} = \tilde{g} - K\gamma \tilde{\rho}^{\gamma-2} \frac{\partial \tilde{\rho}}{\partial \tilde{r}}$$

B-8 (0.6 pt)

$$\frac{d^2 \tilde{r}}{dt^2} = \ddot{\tilde{r}} = \tilde{g} - K\gamma \tilde{\rho}^{\gamma-2} \frac{\partial \tilde{\rho}}{\partial \tilde{r}}$$

B.9. Using of the results in B.7 and B.8, we have:

$$\begin{aligned} \frac{d^2 \tilde{r}}{dt^2} = \ddot{\tilde{r}} &= \tilde{g} - K\gamma \tilde{\rho}^{\gamma-2} \frac{\partial \tilde{\rho}}{\partial \tilde{r}} = g(1 - 2\epsilon) - K\gamma \rho^{\gamma-2} \frac{\partial \rho}{\partial r} \left(\frac{(1 - 3\epsilon)^{\gamma-1}}{(1 + \epsilon)} \right) \\ &= g(1 - 2\epsilon) - K\gamma \rho^{\gamma-2} \frac{\partial \rho}{\partial r} (1 - 3(\gamma - 1)\epsilon - \epsilon) \end{aligned}$$

Equilibrium requires:

$$g - K\gamma \rho^{\gamma-2} \frac{\partial \rho}{\partial r} = 0 \Rightarrow K\gamma \rho^{\gamma-2} \frac{\partial \rho}{\partial r} = g$$

therefore:

$$\ddot{\tilde{r}} = r\ddot{\epsilon} = g(1 - 2\epsilon) - g(1 - 3(\gamma - 1)\epsilon - \epsilon) = g(3\gamma - 4)\epsilon$$

and:

$$\ddot{\epsilon} = \frac{g}{r}(3\gamma - 4)\epsilon$$

$$\ddot{\epsilon} = -\frac{4\pi G\rho_c}{3}(3\gamma - 4)\epsilon$$

Stability requires that:

$$3\gamma - 4 > 0 \Rightarrow \gamma > \frac{4}{3}$$

and the angular velocity of the oscillations will be:

$$\omega = \sqrt{\frac{4\pi G\rho_c}{3}(3\gamma - 4)}$$

B-9 (0.6 pt)

$$\ddot{\epsilon} = -\frac{4\pi G\rho_c}{3}(3\gamma - 4)\epsilon$$

$$\gamma_{\min} = \frac{4}{3}$$

$$\omega = \sqrt{\frac{4\pi G\rho_c}{3}(3\gamma - 4)}$$